

SECTION A [Multiple choice questions] - 18 marks

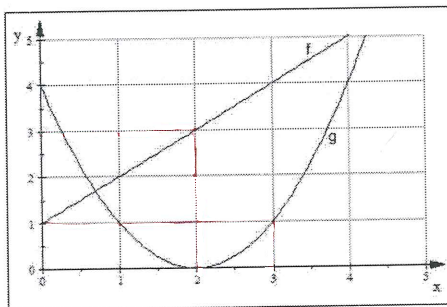
QUESTION 1

If $\sin x = -\frac{1}{3}$, $\pi < x < \frac{3\pi}{2}$, then $\cot x =$

(a) $\frac{1}{\sqrt{8}}$	(b) $-\frac{1}{\sqrt{8}}$	(c) $\frac{3}{\sqrt{8}}$	(d) $-\frac{3}{\sqrt{8}}$	(e) $\sqrt{8}$	(f) $-\sqrt{8}$	(g) None of (a) - (f)
--------------------------	---------------------------	--------------------------	---------------------------	----------------	-----------------	-----------------------

QUESTION 2

Consider the graphs of a straight line f and a parabola g below.



$(g \circ f)(2) =$

(a) 0	(b) 1	(c) 2	(d) 3	(e) 4	(f) 5	(g) None of (a) - (f)
-------	-------	-------	-------	-------	-------	-----------------------

QUESTION 3

If $2e^{x+\ln 3} = 3$, then $x =$

(a) $\ln \frac{1}{2}$	(b) $\ln 2$	(c) $\ln \frac{1}{3}$	(d) $\ln 3$	(e) $\ln \frac{2}{3}$	(f) $\ln \frac{3}{2}$	(g) None of (a) - (f)
-----------------------	-------------	-----------------------	-------------	-----------------------	-----------------------	-----------------------

QUESTION 4

If $f(x) = \ln 2 - \ln x$, then $f^{-1}(x) =$

(a) e^{2-x}	(b) e^{x-2}	(c) $e^{\frac{x}{2}}$	(d) $e^{\frac{x}{2}}$	(e) $\frac{2}{e^x}$	(f) $\frac{e^x}{2}$	(g) None of (a) - (f)
---------------	---------------	-----------------------	-----------------------	---------------------	---------------------	-----------------------

QUESTION 5

For which value(s) of x is $y = \cos^{-1}(1 + 2x)$ defined?

(a) $x \in [-1, 1]$	(b) $x \in [-\frac{1}{2}, \frac{\pi}{2}]$	(c) $x \in [-1, 0]$	(d) $x \in [0, 1]$	(e) $x \in [0, \pi]$	(f) $x \in \mathbb{R}$
---------------------	---	---------------------	--------------------	----------------------	------------------------

QUESTION 6

The relationship between x and y is given by $y + 3x = 10$. What interval on the y -axis corresponds to the interval $x \in [-20, 50]$ on the x -axis?

(a) $y \in [-140, -70]$	(b) $y \in [-140, 70]$	(c) $y \in [-70, 140]$	(d) $y \in [-50, 160]$	(e) None of (a) - (d)
-------------------------	------------------------	------------------------	------------------------	-----------------------

QUESTION 7

Consider the following statements:

$$(1) \cos \frac{3\pi}{4} + \sin \frac{10\pi}{3} = -\cos \frac{\pi}{4} + \sin \frac{4\pi}{3}.$$

$$(2) \cos \frac{3\pi}{4} + \sin \frac{10\pi}{3} = -\cos \frac{\pi}{4} + \sin \frac{\pi}{3}.$$

$$(3) \cos \frac{\pi}{4} + \sin \frac{\pi}{4} = \sqrt{2}.$$

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

ROUGH WORKING

QUESTION 8

The domain of the function $f(x) = \frac{\ln(x-3)}{1 + \ln(x+1)}$ is

- (a) $(-1, \infty)$.
- (b) $(3, \infty)$.
- (c) $x \in \mathbb{R}, x \neq -1 + \frac{1}{e}$.
- (d) $(-1 + \frac{1}{e}, \infty)$.
- (e) $(-1, -1 + \frac{1}{e}) \cup (-1 + \frac{1}{e}, 3) \cup (3, \infty)$.
- (f) None of these.

QUESTION 9

Consider $F(x) = \sec^4(x-4)$. If $F(x) = (h \circ g \circ f)(x)$, then the functions f, g and h can be represented as follows:

- (a) $f(x) = \sec x, g(x) = x-4, h(x) = x^4$.
- (b) $f(x) = \sec x, g(x) = x^4, h(x) = x-4$.
- (c) $f(x) = x-4, g(x) = \sec x, h(x) = x^4$.
- (d) $f(x) = x-4, g(x) = x^4, h(x) = \sec x$.
- (e) $f(x) = x^4, g(x) = \sec x, h(x) = x-4$.
- (f) $f(x) = x^4, g(x) = x-4, h(x) = \sec x$.

QUESTION 10

Consider the following statements:

- (1) $\ln a^2 b^2 = 2 \ln a \ln b$.
- (2) $\ln e^3 = 3$.
- (3) $\ln(1 + e^x) = 0$.

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

QUESTION 11

What is the smallest number that would make $792 \times n$ a perfect square?

(a) 2	(b) 10	(c) 11	(d) 20	(e) 22	(f) None of (a) - (e)
-------	--------	--------	--------	--------	-----------------------

QUESTION 12

The equation of the graph obtained by shifting the graph of $y = f(x)$ to the right by three units, reflecting it about the x -axis and stretching it vertically by a factor of 2, is given by:

- (a) $-2f(x + 3)$.
- (b) $2f(x + 3)$.
- (c) $\frac{1}{2}f(x - 3)$.
- (d) $-2f(x - 3)$.
- (e) $-2(f(x) - 3)$.
- (f) $\frac{1}{2}(f(x) - 3)$.
- (g) None of these.

End of Section A

ROUGH WORKING

SECTION B [Fill in questions] - 8 marks

SECTIONS B AND C MUST BE SCANNED AND UPLOADED ON CLICKUP.
Please do not scan your rough working or any empty pages.

SURNAME:

INITIALS:

STUDENT NUMBER:

SECTION B: Write only your final, fully simplified answer in the block provided. Only this will be marked.

QUESTION 13 [1 mark]

Assume that $a < 0$ and $b > 0$. Describe the set in which x lies if x is less than or equal to a and greater than b .

QUESTION 14 [1 mark]

Write down the equation of the function $f(x) = Be^x$ that passes through the point $(1, 2)$. Fully simplify your answer.

QUESTION 15 [1 mark]

If $\frac{(e^{2x})^3 e^{3x}}{e^{4x}} = e^a$, then $a =$

QUESTION 16 [1 mark]

Solve the inequality $\ln(x^2) - \ln 6 > 0$. Write down your answer using interval notation.

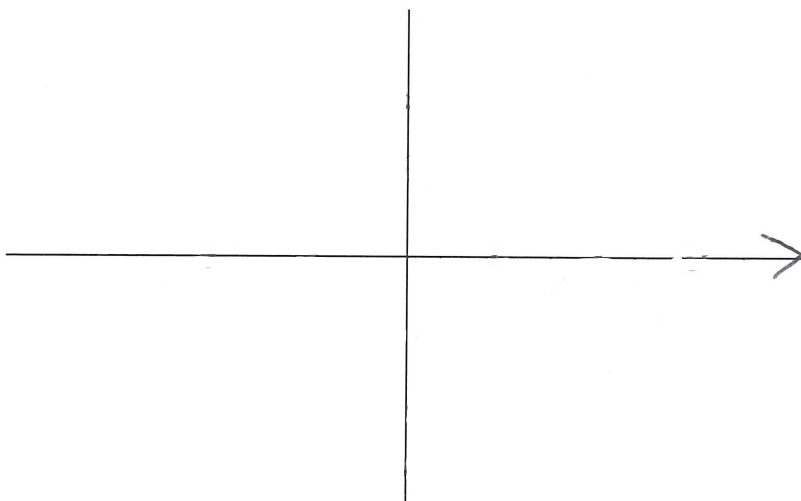
QUESTION 17 [1 mark]

Solve the inequality $\frac{|x|}{x-2} < 0$. Write down your answer using interval notation.

QUESTION 18 [3 marks]

$$\text{Let } f(x) = \begin{cases} 1 + \frac{x}{2} & \text{if } -2 \leq x < 0, \\ x - 3 & \text{if } 0 \leq x < 3. \end{cases}$$

- (i) Sketch the graph of the function f on the axes provided, indicating all relevant intercepts and information. (1)



- (ii) Determine the domain of f^{-1} . (1)

- (iii) Determine $f^{-1}(-1)$. (1)

End of Section B

ROUGH WORKING

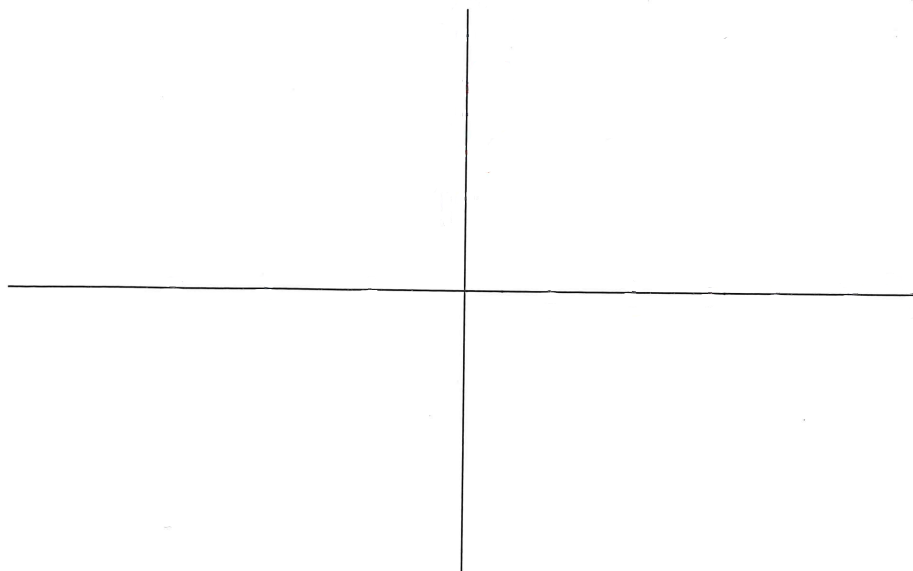
SECTION C [Full written answers] - 24 marks

QUESTION 19 [5 marks]

(a) Give the definition of the absolute value of a real number x . (1)

(b) Use the definition of the absolute value to rewrite $\sqrt{|x-1|+4}$ as a piecewise defined function. Simplify completely. (2)

(c) Sketch the graph of the function $\sqrt{|x-1|+4}$ on the axes provided, indicating all relevant intercepts and information. (2)



QUESTION 20 [7 marks]

Solve for x . Write your answer in interval notation.

(a) $\ln(x - 1) + \ln(5 - x) < \ln 3$ (4)

(b) $8e^{-x} > \frac{1}{e}$ (3)

QUESTION 21 [4 marks]

- (a) Complete the statement (note that $\tan^{-1} x = \arctan x$): (1)

$\tan^{-1} x = y$ if and only if

- (b) Determine the value of (3)

$$\sin(2 \tan^{-1} x).$$

Explain all your steps.

QUESTION 22 [3 marks]

Consider the function

$$f(x) = \frac{x-2}{x+3}.$$

Find the inverse function, f^{-1} , together with the domain and range of f^{-1} . Use interval notation.

QUESTION 23 [3 marks]

(a) Write down the definition of an odd function. (1)

(b) Assume that f is an odd function with domain $\mathbb{R}$. Is the composite function $f \circ f$ also an odd function? Show all calculations and explain your answer. (2)

QUESTION 24 [2 marks]

Consider the functions

$$f(x) = a^x, a > 0, \text{ and } g(x) = e^{kx}, k \in \mathbb{R}.$$

Given that $f(x) = g(x)$, in which interval does k lie if $a > 2$.

ROUGH WORKING
